# Biliary Atresia, Part Nine: Portal Hypertension after the Kasai Operation. Varices in Infants, the Debate over Primary Prophylaxis, and Hypersplenism.

Imagine an eleven-month-old girl who had a Kasai operation for biliary atresia at forty-eight days of age. The operation worked: her jaundice cleared within three months, her stools are yellow and she is growing. At this visit you feel her spleen three centimetres below the left costal margin, and her platelet count is one hundred and sixty-eight thousand per microlitre, which is still within the normal range. Her parents ask whether the enlarged spleen matters. You find yourself facing three questions at once. Should she have an endoscopy now, at less than a year old? If varices are found, should they be treated before they ever bleed? And what, if anything, should be done about the spleen itself?

The main idea of this chapter is simple to state and harder to apply. Portal hypertension is the main long-term threat to a native liver that has been drained successfully by the Kasai. Varices can appear early, sometimes within the first year of life, and in infants they can hide behind a perfectly normal platelet count. A first variceal bleed is uncommon and, in modern series, rarely fatal, so the case for treating varices before they bleed rests mainly on avoiding morbidity rather than on preventing death. Experienced centres scope children who have signs of portal hypertension and treat high-risk varices endoscopically when transplantation is not close, but the evidence for this approach is observational, because no paediatric randomised trial has settled the question. Finally, a variceal bleed is not only an emergency. It is a prognostic event, and it should start a conversation about liver transplantation.

Some terms first. Portal hypertension means raised blood pressure in the portal vein, the vessel that carries blood from the intestines and spleen to the liver. When the pressure rises, blood looks for other routes back to the heart. These detours are called collaterals, and the most dangerous of them are varices, which are dilated, thin-walled veins in the lower oesophagus and the upper stomach that can rupture and bleed. Hypersplenism means an enlarged, congested spleen that traps and destroys blood cells, so that the platelet count, and sometimes the white cell count, falls. Non-selective beta-blockers, such as propranolol and carvedilol, are drugs that lower portal pressure by reducing the heart's output and by narrowing the blood vessels of the gut. Endoscopic band ligation means placing small rubber bands around varices through an endoscope so that they clot and scar. Sclerotherapy means injecting an irritant solution into or beside a varix for the same purpose.

The evidence in this chapter comes from a handful of key sources. There is the North American Childhood Liver Disease Research Network, known as ChiLDReN, which described

portal hypertension in a cross-sectional study led by Shneider and published in the Journal of Pediatric Gastroenterology and Nutrition in 2012, and which followed a much larger group of children over time in a study led by Bass and published in Hepatology in 2022. There is a long series of papers from the Bicêtre hospital in Paris, led by Duché and Ackermann, the most recent of which appeared in Gastroenterology in 2026. There is the Japanese Biliary Atresia Registry, analysed for portal hypertension by Okubo and Nio in 2025. And there are the documents that shape practice: the 2016 summary of the paediatric satellite symposium of the Baveno Six consensus, the adult Baveno Seven consensus of 2022, the adult practice guidance of the American Association for the Study of Liver Diseases published in 2024, and a 2026 statement on endoscopic management of variceal bleeding from the Asian Pan-Pacific Society for Pediatric Gastroenterology, Hepatology and Nutrition.

## Part one. Why a drained liver still develops portal hypertension.

It may seem odd that a child whose jaundice has cleared can go on to develop portal hypertension. The explanation is that the Kasai operation treats only part of the disease. It bypasses the blocked bile duct outside the liver, but biliary atresia is also an inflammatory, scarring disease of the small bile ducts inside the liver, and that process continues after the operation. Episodes of cholangitis, which is infection of the bile ducts, add further injury. Over months and years, scar tissue builds up around the portal tracts and the liver develops regenerative nodules. Both changes increase the resistance to blood flowing into the liver. At the same time, blood flow into the portal system from the gut and spleen tends to increase. Higher resistance and higher inflow together raise portal pressure.

You can think of the consequences as a chain. First, the spleen becomes congested and enlarges, and because an enlarged spleen holds back platelets and white cells, the blood counts fall. Next, collateral veins open up to decompress the portal system. The important ones are in the lower oesophagus and around the top of the stomach, but in children who have had a Kasai there is also a special site: the loop of small bowel that was joined to the liver, called the Roux loop, can develop varices where it meets the liver surface. Finally come the clinical events: bleeding from varices, ascites, which is fluid in the abdomen, and the two lung complications, hepatopulmonary syndrome and portopulmonary hypertension. Each of these events worsens the outlook for the native liver.

Infants differ from adults in two important ways. The first is timing. In many infants, fibrosis is already advanced at the time of the Kasai, so portal hypertension can be established within months, when the child weighs less than ten kilograms and the oesophagus is small. The second is the markers. In adults, a palpable spleen and a platelet count below one hundred and fifty thousand are useful clues to portal hypertension. In infants the spleen is small to begin with and platelet counts are normally high, so these clues appear late. And in children we do

not routinely measure portal pressure directly. There is no everyday paediatric equivalent of the hepatic venous pressure gradient, the catheter measurement that defines clinically significant portal hypertension in adults.

## Part two. What we mean by portal hypertension, and how common it is.

Because pressure is not measured, paediatric studies use clinical definitions. The ChiLDReN network defined definite portal hypertension as either a past complication of portal hypertension, such as a variceal bleed or ascites, or the combination of a spleen palpable more than two centimetres below the costal margin together with a platelet count below one hundred and fifty thousand. Possible portal hypertension meant only one of those two findings. The Paris group introduced a different and more practical concept, high-risk varices. Varices are high-risk if they are large, called grade three; or if they are medium-sized, grade two, with red marks on their surface, called red wale signs, which indicate thin walls under strain; or if there are varices in the stomach extending along the cardia. In their 2017 paper, high-risk varices were present in ninety-six percent of the children who had bled spontaneously, compared with eleven percent of the children who never bled. In 2026, the same group refined this into a score from zero to five, with three to five counting as high risk. When you read any paediatric series, check whether portal hypertension means a clinical sign, an endoscopic finding or an event, because the three give very different prevalences.

With those definitions in mind, how common is it? In the ChiLDReN cross-sectional study of one hundred and sixty-three children who were living with their native liver, at an average age of about nine years, forty-nine percent had definite portal hypertension, seventeen percent had possible portal hypertension, and only thirty-four percent had none. About one child in five had already had a variceal bleed. Only eight of the one hundred and sixty-three were taking beta-blockers. And about a third of those with definite portal hypertension also had signs of reduced liver function, either a prolonged prothrombin time or a low albumin. In the Japanese registry of three thousand seven hundred and seventy-seven patients followed from 1989 to 2021, portal hypertension was recorded at some point in forty-six percent, varices in thirty-five percent and hypersplenism in thirty-seven percent. Varices needed treatment in about fifteen percent, and hypersplenism in about six percent. Among children who reached twenty years with their own liver, half had portal hypertension, mostly diagnosed between the second and the fifteenth year. The lesson is that portal hypertension is the rule, not the exception, in long-term native-liver survivors, but that only a minority need treatment for it.

## Part three. How often children bleed, and when.

The strongest data on bleeding risk come from the ChiLDReN longitudinal study by Bass and colleagues, which followed eight hundred and sixty-nine children. It had two groups. The first, called PROBE, enrolled five hundred and twenty-one infants within the first few months after

their Kasai. The second, called BASIC, enrolled three hundred and forty-eight children who were already older than three years. In the infant group, the chance of a first variceal bleed by five years was about nine percent, with a ninety-five percent confidence interval of seven to twelve percent. In the older group it was eight percent. Baseline features that predicted bleeding were the signs of portal hypertension themselves, a low platelet count, a higher bilirubin, a higher liver enzyme level, a lower albumin, and a higher ratio of the liver enzyme to platelets. Eighty children bled during the study. Of these, two died, which is two and a half percent, and ten, which is twelve and a half percent, received a liver transplant within six weeks. The authors concluded that this low risk of bleeding and low mortality should inform decisions about screening and about primary prophylaxis. In other words, most children never bleed, and few die when they do. That is the strongest argument against universal endoscopic surveillance.

The French experience gives a different emphasis. In the 2010 Paris study of one hundred and thirty-nine children with biliary atresia, one hundred and twenty-five had signs of portal hypertension and were scoped at a median age of thirteen months. Seventy percent of them already had varices. Twenty percent bled, at a median age of seventeen months, and three quarters of the bleeds happened before two and a half years of age. The finding at first endoscopy predicted the future. Over five years, the risk of bleeding was fifty-nine percent in children with grade three varices, seventeen percent with grade two varices and twelve percent with no varices at all. Gastric varices, red wale signs and changes in the stomach lining called portal hypertensive gastropathy also predicted bleeding. The message is that in biliary atresia the window for high-risk varices opens in the first two years, precisely when endoscopy is technically hardest. But remember that these data come from a single expert centre with a policy of early surveillance. They show what endoscopy finds; they do not show that finding it improves survival.

The 2026 Paris study by Ackermann and colleagues adds the natural history of the varices themselves. It included one thousand five hundred and eighty-six children with portal hypertension, of whom five hundred and ninety had at least two endoscopies, and four hundred and three of those had biliary atresia. Over a mean of four years, the endoscopic picture worsened in fifty-eight percent. In children who started without high-risk varices, the probability of developing them was thirty-six percent at five years and fifty-four percent at ten years. Bleeding occurred in under four percent of children without high-risk varices at their last endoscopy, but in seventy-five percent of children who had high-risk varices and were not given endoscopic or surgical prophylaxis. The most important finding for clinical practice concerns platelets. Of all children with high-risk varices, about a third had a platelet count of one hundred and fifty thousand or more. In infants younger than twelve months, that proportion was sixty-two percent, falling to just two percent in children older than ten years.

So a normal platelet count in a ten-month-old does not reassure you. The authors went as far as proposing that infants with biliary atresia under twelve months of age should have grade two varices or gastric varices along the lesser curve treated prophylactically.

Bleeding is not only a problem of infancy. In the ChiLDReN cross-sectional study, among the thirty-two children who had bled, seven first bled before two years of age, seven between two and five, eleven between six and ten, and seven after ten. This is a survivor cohort, so children who bled early and then died or were transplanted are under-represented, but it shows that bleeding risk is spread across childhood. Early bleeds come from infants with the worst fibrosis; late bleeds come from slowly progressive fibrosis in children who looked well for years. The King's College Hospital study of adolescents, published by Jain and colleagues in the Journal of Hepatology in 2019, found that the presence of varices by age sixteen increased the hazard of needing a transplant in adulthood more than eightfold, and that portal hypertension by age sixteen increased it about sevenfold. Surveillance is therefore lifelong, and the same child needs a different question at one, six and sixteen years of age.

## Part four. Can we avoid the endoscope?

Because endoscopy in a small child means a general anaesthetic, much effort has gone into non-invasive tests that could predict varices. The simplest is the platelet count. A Cochrane diagnostic review by Colli, Gana and colleagues in 2017 pooled four paediatric studies with two hundred and seventy-seven children and found a sensitivity of about seventy-one percent and a specificity of about eighty-three percent for varices. As we have just seen, platelets perform worst in infants. A refinement is to divide the platelet count by the spleen size expressed as a z-score, which adjusts for age. In a multicentre validation study of one hundred and eight children led by Gana and published in Gastroenterology in 2011, this ratio had an area under the curve of zero point eight four. The same group derived a clinical prediction rule that combines platelets, spleen size z-score and albumin. In the original derivation cohort its area under the curve was zero point nine three, but in the validation cohort it was zero point eight. Liver stiffness measured by transient elastography has also been studied in biliary atresia. In a study from Thailand by Chongsrisawat and colleagues, a stiffness cut-off of twelve point seven kilopascals had an area under the curve of zero point eight nine, with a sensitivity of eighty-four percent and a specificity of seventy-seven percent, and the ratio of liver enzyme to platelets performed nearly as well. Spleen stiffness measured by magnetic resonance elastography looked promising in a very small Korean study of twenty-two children.

The common thread is that these tests are reasonably good at telling you that a child probably has varices of some size. They are poor at excluding high-risk varices, and that is the question that matters for prophylaxis. The adult Baveno rule, which allows adults with a liver stiffness below twenty kilopascals and platelets above one hundred and fifty thousand to skip screening

endoscopy, has not been validated in infants. So use non-invasive tests to decide who should be scoped sooner, not who can safely be left alone.

Portal hypertension also carries long-term prognostic weight. In the Japanese registry, among five hundred and ninety-six children whose jaundice had cleared, native-liver survival at thirty years was eighty-eight percent if they never had portal hypertension, sixty-eight percent if they had portal hypertension that did not need treatment, and only forty-six percent if their portal hypertension needed treatment. Treating varices does not cure the liver; it buys time. That is why transplant assessment should be planned alongside endoscopic therapy, not after it fails.

## Part five. The debate about primary prophylaxis.

Primary prophylaxis means treating varices that have never bled, to prevent the first bleed. In adults with cirrhosis this is standard. In children it remains contested, and it helps to lay the arguments side by side.

The argument in favour comes mainly from Paris. High-risk varices were present in almost every child who bled. Three quarters of children with untreated high-risk varices eventually bled. And in their 2013 series of thirty-six children with biliary atresia who received endoscopic primary prophylaxis, at a mean age of twenty-two months, no child had a breakthrough bleed and three-year survival was ninety-seven percent. There is also an argument from morbidity. A variceal bleed in a toddler means emergency transfusion, intensive care, sometimes decompensation of the liver, and a frightened family. In ChiLDReN, one in eight children who bled was transplanted within six weeks.

The argument against comes mainly from North America. The five-year incidence of a first bleed is only about nine percent, and the mortality of a bleed in modern care is about two and a half percent. Treatment is not trivial: in the 2013 Paris series it took an average of just over four sessions to eradicate varices, and varices came back in thirty-seven percent. Each session needs an anaesthetic, and banding and sclerotherapy can cause ulcers and strictures. The 2016 paediatric Baveno symposium therefore explicitly reached no consensus on primary prophylaxis, reasoning that a first bleed appears rarely fatal and that its morbidity is poorly characterised.

Behind both positions lies a thin evidence base. Cochrane reviews led by Cifuentes and Gana in 2021 found a single small feasibility trial comparing band ligation with a sham procedure in twenty-two children. One of twelve children who were banded bled, compared with three of ten in the sham group, giving a relative risk of zero point two eight with a ninety-five percent confidence interval from zero point zero three to two point two seven, which is compatible with both large benefit and no benefit at all. The certainty of this evidence was rated very low. There are no randomised trials of beta-blockers against placebo in children, none comparing

banding with beta-blockers, and the review of sclerotherapy is still at the protocol stage. Every recommendation on primary prophylaxis in children is therefore extrapolated, either from adults or from observational paediatric cohorts. And the adult beta-blocker data, now dominated by carvedilol, rest on a reduction in portal pressure that cannot be measured in a toddler.

The real disagreement is smaller than it looks. Almost nobody bands small varices, and almost everybody treats high-risk varices when transplantation is far away. The practical questions are whether to scope at all in a child without clinical signs, and what to do for a high-risk infant who weighs less than ten kilograms.

## Part six. The tools available in small children.

For band ligation, the bands are mounted on a cap that fits an adult-diameter endoscope, which limits its use in small infants. The 2026 statement from the Asian Pan-Pacific society recommends band ligation as the preferred method from about ten kilograms, and for children under ten kilograms suggests considering sclerotherapy or a non-selective beta-blocker case by case. An Italian survey of sixteen centres published in 2025 found that sixty-three percent used sclerotherapy in children under ten kilograms, that all treated high-risk varices, and that all used propranolol as their beta-blocker of choice, mostly with the dose adjusted according to heart rate. For gastric varices of certain types, injection of a tissue glue called cyanoacrylate is used in expert hands.

Two practical points about beta-blockers in toddlers are worth emphasising. First, a beta-blocker blunts the rise in heart rate that would normally signal a bleed, so a child can look deceptively stable. Second, propranolol can mask the warning signs of low blood sugar, which matters in a small child who may be eating poorly. Families need a clear written action plan.

The Paris 2017 data show where endoscopic prophylaxis leads. Among one hundred and twenty children with cirrhosis and high-risk varices, one hundred and ten had endoscopic treatment. Varices were eradicated in seventy-six, and twenty of those relapsed. Of the thirty-four children in whom varices could not be eradicated, twenty-nine were transplanted. Endoscopic prophylaxis is a bridge, and the bridge often leads to transplantation.

## Part seven. Managing an acute variceal bleed.

When a child with biliary atresia vomits blood or passes black stools, the principles are the same as in adults, with some paediatric adjustments. The first step is resuscitation with a restrictive approach to transfusion. The 2026 Asian Pan-Pacific statement advises transfusing red cells when haemoglobin falls below seven grams per decilitre, with a target of seven to nine, and advises against routine fresh frozen plasma. Over-transfusion raises portal pressure

and increases rebleeding, and children are no exception. The second step is a drug that lowers portal blood flow. In children this is usually octreotide, given as a bolus of one microgram per kilogram, to a maximum of fifty micrograms, followed by an infusion of one to two micrograms per kilogram per hour for two to five days. The third step is antibiotics, typically a third-generation cephalosporin for five to seven days, because infection is common after a variceal bleed and worsens outcome. The fourth step is endoscopy, within twenty-four hours of resuscitation in the paediatric statement, compared with within twelve hours in the adult Baveno and American guidance. Band ligation is preferred over sclerotherapy where the child's size allows. After the bleed comes secondary prophylaxis, meaning repeated banding every two to four weeks until the varices are gone.

One adult intervention rarely transfers. In adults at high risk of rebleeding, Baveno Seven recommends early placement of a transjugular intrahepatic portosystemic shunt, known as a TIPS. In a small child with biliary atresia, whose portal vein is often small, this is rarely available or appropriate, although a meta-analysis of paediatric TIPS reported technical success in ninety-four percent, with shunt dysfunction in twenty-seven percent and encephalopathy in about eleven percent. In biliary atresia, TIPS is mainly a bridge to transplantation. And above all, remember the ChiLDReN numbers: two and a half percent died and twelve and a half percent were transplanted within six weeks of a bleed. A bleed is the moment to recalculate the prognosis, not only to stop the bleeding.

## Part eight. Hypersplenism and other bleeding sites.

Hypersplenism is common, recorded in about thirty-seven percent of the Japanese registry, but it was treated in only six percent. In most children it is a marker of portal hypertension rather than a target for treatment. A low platelet count alone is rarely a reason to intervene. Intervention is considered when thrombocytopenia contributes to bleeding, or when a very large spleen limits normal activity or sport. The options are partial splenic embolisation, in which a radiologist blocks part of the splenic arterial supply, and partial splenectomy. In a small 2026 series of twelve children, partial embolisation raised the median platelet count from about fifty-eight thousand by a further two hundred and thirty-five thousand, with abdominal pain in a third and fever in about a sixth, but no major complications. In a Japanese series of fifteen children with biliary atresia who had partial splenectomy, platelets remained higher at three years, but about a quarter later needed a living-donor transplant. Total splenectomy should be avoided: it carries a lifelong risk of overwhelming infection, removes a natural route by which portal blood escapes, and can complicate later transplantation.

There is also a bleeding source that is easy to miss. If a child with biliary atresia has black stools but the oesophagus and stomach look clean at endoscopy, think of varices at the junction of the Roux loop and the liver, or of portal hypertensive changes in the small bowel. In a 2024

Korean series of over three hundred children scoped after the Kasai, such enteropathy was found in about fifteen percent, and in almost a third of them it was the only source of bleeding. Capsule endoscopy, enteroscopy or angiography may be needed.

## Part nine. What the guidelines say.

There is no guideline on portal hypertension from the European or North American paediatric gastroenterology societies. The documents that exist are adult guidance and paediatric expert statements, and comparing them shows where children differ.

On screening endoscopy, the 2024 American guidance and Baveno Seven both allow adults with favourable stiffness and platelet values to avoid screening endoscopy. Paediatric experts state that non-invasive tests cannot replace endoscopy, and the Paris platelet data in infants support them. This is a material difference. On primary prophylaxis, adults are offered a non-selective beta-blocker, with carvedilol preferred, or band ligation. Paediatric experts reached no consensus, and those who treat use banding in children over ten kilograms. This too is a material difference, because the adult beta-blocker-first strategy has no paediatric evidence. On secondary prophylaxis after a bleed, adults receive a beta-blocker combined with banding, while paediatric practice relies mainly on banding, so the two differ. On acute bleeding, both use vasoactive drugs, antibiotics and endoscopy, but the paediatric timing is more permissive and pre-emptive TIPS is rarely relevant, so they differ in detail. On restrictive transfusion, all are aligned. And on hypersplenism, no document gives guidance at all, which is a gap. The 2024 American guidance states explicitly that it applies only to adults.

## Part ten. Traps, pearls, and what to do in clinic.

Several traps recur in practice. The first is being reassured by a normal platelet count in an infant, when sixty-two percent of infants with high-risk varices have one. The second is applying adult elastography cut-offs or adult carvedilol strategies to toddlers without evidence. The third is over-transfusing a bleeding child, which raises portal pressure. The fourth is removing the spleen to treat a number. The fifth is missing Roux-loop varices when the upper endoscopy is clean. But the biggest trap is treating varices as a purely local problem. A child who needs repeated banding usually has a failing native liver, and in Japan fewer than half of jaundice-free children with treated portal hypertension kept their liver to thirty years. Every endoscopy should come with the question: is it time to list this child for transplantation?

To pull the numbers together: definite portal hypertension was present in forty-nine percent of North American native-liver survivors at an average age of nine, and was recorded at some point in forty-six percent of Japanese patients. The first variceal bleed occurs in about nine percent within five years of the Kasai; after a bleed, about two and a half percent die and twelve and a half percent are transplanted within six weeks. In Paris, seventy percent of scoped

infants had varices at a median of thirteen months, and the five-year bleeding risk was fifty-nine percent with grade three varices against twelve percent without varices. High-risk varices are found in ninety-six percent of children who bleed. Sixty-two percent of infants under one year with high-risk varices have normal platelets. There is one small paediatric trial of banding and none of beta-blockers. Banding becomes practical at around ten kilograms. In an acute bleed, transfuse below a haemoglobin of seven, give octreotide and antibiotics, and scope within a day. And treated portal hypertension is a transplant signal, with thirty-year native-liver survival of forty-six percent against eighty-eight percent without portal hypertension.

How would you manage the five common situations? For the eleven-month-old girl from the start of this chapter, with a cleared bilirubin, a spleen three centimetres below the costal margin and normal platelets, arrange an endoscopy in an experienced unit, because platelets mislead in infants and high-risk varices can appear early. For an eight-kilogram infant with grade three varices and red signs but good liver function, treat with sclerotherapy or a beta-blocker now and move to banding once the child reaches ten kilograms, because transplantation is not close and untreated high-risk varices bleed often. For a toddler with small grade one varices and good liver function, do not treat, and repeat the endoscopy in one to two years. For a two-year-old with a first bleed and an albumin of thirty grams per litre, give octreotide and antibiotics, keep the haemoglobin between seven and nine, band the varices and start a transplant assessment. And for a teenager with a platelet count of sixty thousand, no varices and a love of sport, offer a spleen guard rather than a splenectomy, and consider partial embolisation only for a clear reason.

What should you do differently in clinic? Measure the spleen and record platelets and albumin at every visit, and do not let a normal platelet count in an infant reassure you. Agree a local policy for endoscopy in the first year for children with signs of portal hypertension, and treat high-risk varices when transplantation is not imminent, choosing the technique by the child's weight. Give every family a written bleeding plan and a letter for the emergency department. Keep surveillance going through adolescence. And whenever a child needs variceal therapy or has a bleed, ask the transplant team to see them, because in biliary atresia portal hypertension is not only a plumbing problem. It is a measure of how much liver is left.